Ali Keramati

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Education

University of California Irvine **Sep. 2024 – Jun. 2029**

Ph.D in **Education Data Science** (GPA: 4/4) **Expected Graduation Date: June 2029** *Irvine, CA, USA*

- **Relevant Courses:** *Neural Networks and Deep Learning, Deep Generative Models, Neural Architecture of Language, Language Models in Cognitive Science, Learning Analytics Practicum, Qualitative Research Methods, Research Epistemologies and Methods, Applied Regression, Brain Inspired Learning Algorithms*

University of Tehran **Sep. 2019 – Jul. 2024**

B.Sc. in **Computer Engineering** (GPA: 18.51/20) + Minor in **Psychology and Education** (GPA: 19.21/20) *Tehran, Iran*

- **Major Degree Courses:** *Discrete Mathematics, Data Structures and Algorithms, Artificial Intelligence, Machines and Language Theory, Engineering Probability and Statistics, Internet Engineering, Foundations of IT, Large Language Models*
- **Minor Degree Courses:** *Educational Psychology, Instructional Design, Curriculum Development, Intro to Psychological Tests, Teaching Methods, Qualitative and Quantitative Research Methods, Philosophical Schools and Educational Theories*

Research Interests

- Multi-Agent and **Agentic AI**: Tool-use/RAG/Memory
- **HCI**: Human-Centered AI/Human-in-the-Loop AI
- **LLM-as-Judge**: Automated Evaluation/Benchmarking
- **Fine-Tuning** & Alignment: LoRA/Adapters/RLxF
- **RL**: MDPs/Reward Modeling/Constrained/Actor–Critic
- Robust, Safe, Explainable, Trustworthy, Responsible AI
- Personalized Recommender Systems & Conversational-AI
- Cognitive User Modeling & Human Behavior Analysis

Work Experience

Research Scientist Intern at Harvard University **Jun. 2025 - Sep. 2025**

Child-Centered AI Lab, PI: Prof. Y. Xu *Cambridge, MA*

- Developed Curio 2, a LangGraph-based science chatbot that utilizes graph-based dialogue and state management for enhanced Transparency and Explainability **(XAI)**, while promoting **Trustworthy AI** through Child-Safe design, robust dialogue management, and safeguards against harmful content. System alignment was achieved via a self-correction multi-agentic pipeline grounded in **Constitutional AI**, with performance validated against K–12 standards using a multi-LLM-as-judge evaluation.
- Compared two voice interaction architectures, **realtime (speech-to-speech)** and **chained**, to evaluate trade-offs between latency, transparency, pedagogical control, and age-appropriate conversational learning experiences in educational voice agents.

Applied Sciences Intern **Jun. 2026 - Sep. 2026**

Microsoft *Redmond, WA*

- Incoming intern; team and project assignment pending.

AI/ML Engineer **Sep. 2022 - Jul. 2024**

Tadvin Farayand Co. *Tehran, Iran*

- Analyzed structured ERP and unstructured industrial data; designed (NLP+RL)-driven workflows to enable autonomous collaboration across ERP modules for inventory analysis, document processing, and predictive maintenance.

Selected Publications

[1] Keramati A., Zhou S., Mehrotra S., and Warschauer M. **"From Advocacy to Judgment: Training-Free Analytic Scoring with Multi-Agent Debate and Exemplar Retrieval"**, (ACL 2026)

[2] Keramati A., Cheok J., Horne J. and Warschauer M. **"The First Tokens Matter: Early Confidence Signals for Evaluating LLM Reasoning"** (ICLR 2026)

[3] Keramati A. and Warschauer M. **"Application of Multi-Agent Systems for Essay Scoring"**, NeurIPS 2025

[4] Keramati A., Jie C., Mohammadi I., Warschauer M. and Shi Y. (in press) **"Simulating Students' Java Programming Errors with Large Language Models"** Nineteenth International Conference on Educational Data Mining (EDM 2026)

[5] Tajik E., Shahrokhian B., Borchers C., Keramati A., Simon S., Pal S. and Sankaranarayanan S. **"Disagreement as Data: Reasoning Trace Analytics in Multi-Agent systems"**, (LAK 2026)

[6] Mohamadi I., Keramati A. and Rismanchian A. (in press) **"Mapping Instructors' Attitudes Towards Cheating with AI: An Exploratory Analysis Using Forum Analysis"**, Computer-Supported Collaborative Learning (CSCL 2026)

[7] Fallah A., Keramati A., Nazari M., and Mirfazeli F. **"Automating Theory of Mind Assessment with a LLaMA-3-Powered Chatbot: Enhancing Faux Pas Detection in Autism"**, *(ICCKE 2024)*

[8] Shahhoseini N., Taghiyareh F., and Keramati A. **"iTAG: Easy, Rapid, Automatic Intelligent Tagging for Educational Contents"**, *2025 29th International Computer Conference, Computer Society of Iran (CSICC 2025)*

[9] Keramati A., Keramati M. R., and Arefian M. H. **Students' Reflection on the Effect of Collaborative Learning on Learning Environment and Academic Achievement in Online Reflective Platforms**, *Reflective Practice Journal*

[10] Vatandoust M., Mohajeri M. M., Keramati A., and Ahmadabadi M. N. **"AI-powered Digital Framework for Personalized Economical Quality Learning at Scale"**, (arXiv, 2025).

Honors & Awards

NeurIPS 2025 — AI Education Workshop Program **Dec. 2025**
Accepted Oral Presentation *San Diego, CA*

Scholarship Recipient, LearnLab Summer School at Carnegie Mellon University **Jul. 2025**
Awarded funding to attend the program on Computer Science Education Research track at CMU *Pittsburgh, PA*

Graduate Ambassador Candidate – TPC Mentor Fellowship at *UC Irvine* **Jun. 2025**
Chosen to oversee mentor–mentee coordination and selection, enhancing communication and program engagement *Irvine, CA*

Top 10 — B.Sc. Computer Engineering (2019 cohort) at *University of Tehran* **Jul. 2024**
Ranked within the top 10 of the graduating cohort based on cumulative GPA *Tehran, Iran*

Top 0.1%, Konkour — National University Entrance Exam **Aug. 2019**
Ranked among hundreds of thousands of examinees; admitted to B.Sc. in CE at Iran's top-ranked university *Iran*

Technical Skills

Programming Language: Python, C/C++/C#, Java, Golang, JavaScript (Node.js, React), SQL, HTML/CSS, Verilog
Agentic Frameworks: LangChain, LangGraph, AutoGen, CrewAI, Swarm (OpenAI), Mem0AI, ReAct, Haystack
DL Fine-Tuning & Alignment: PyTorch, Hugging Face Transformers, PEFT, TRL, OpenDelta, Diffusers, vLLM
Retrieval & Memory: FAISS, ChromaDB, Neo4j, Weaviate, Elasticsearch, Redis, PostgreSQL, MongoDB
Data Analytics: Pandas, NumPy, Scikit-learn, Spark, R, Power BI, LearnSphere, OpenLA, PyBKT, ATLAS.ti, Dedoose
Human-Centered & Cognitive Tools: Unity ML-Agents, PsychoPy, PsyToolkit, OpenAI Agents SDK

Selected Research Projects

University of California, Irvine

- *PapyrusAI:* Designed a GPT-based personalized generative writing coach leveraging an **agentic RL** pipeline capable of autonomous decision-making for tool invocation, web search, and dynamic **RAG** utilization. Integrated fine-tuned **BERT** for labeling argumentative essays and LLaMA-3.1-8B with **LoRA** + 4-bit **quantization** for essay evaluation.
- Uncertainty estimation in multi-agent LLMs, using token-level log-probs as proxies for agent confidence. Comparing first-token logprobs, sequence-level likelihoods, multi-sample voting, and perplexity-based signals to assess reliability.
- Developed an adaptive **few-shot RAG** system with continual memory curation: admission control, candidate→gold promotion, and feedback-driven ranking to keep high-quality exemplars highly retrievable as the database grows.
- As **TA**, developed an **automated grading** and **feedback** system for weekly assignments and co-built an **AI detection RL system** using LLMs to identify AI-generated submissions based on writing patterns, timing, and key concepts.
- Extending VAEs for Collaborative Filtering with modern training strategies on large implicit-feedback datasets, and benchmarking these VAE-based recommenders against language-model–based recommenders trained on the same data.

LearnLab Summer School (2025) at Carnegie Mellon University

- Developed a multi-agent **student simulator + verifier** framework for generating and verifying Java programs with realistic logical errors. Designed a student simulator using *Self-Refine* prompting, and a verifier agent implementing an **RLAIF** loop for structured error assessment and refinement. Evaluated performance via AST- and ZSS-based metrics.
- Analyzing Reddit posts from *r/Teachers* using **LDA** topic modeling, NLP-based classification, and qualitative coding to examine teachers' use of AI tools; training an LLM-based **classifier** to generalize coded themes across large datasets.

University of Tehran

- Developed a multi-agentic chatbot for open learner modeling, using a multi-stage conversational flow to dynamically **profile** and update learner characteristics based on their in-chat interactions. Integrated a **RAG** system that uses a learner's profile as a query to retrieve the top 3 relevant profiles, aligning personalized activity recommendations.
- Automated generation of interactive learning modules from **IEEE LOM metadata** and integrated Avida-ED simulations; applied genetic algorithms to personalize activity parameters and difficulty levels.
- Implemented Theory of Mind assessment for individuals with Autism by developing a **LLaMA-3 chatbot** to administer the Faux Pas Recognition Test. The system uses a multi-phase pipeline that features interactive rehabilitation dialogues with adaptive hints and an automated scoring phase using an **LLM-as-judge** to generate clinician-ready reports.

Certifications

Generative AI with Large Language Models **Jul. 2022**
DeepLearning.AI — Coursera *Online*

Fundamentals of Reinforcement Learning **Aug. 2023**
University of Alberta — Coursera *Online*

LangChain: Chat with Your Data **Jul. 2024**
DeepLearning.AI *Online*

Microsoft Certified: Azure AI Fundamentals **Oct. 2025**
Microsoft Learn *Online*